

Problem Set 1

Industrial Organization

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I. Conduct analysis in the product space: a monopsonist firm?

Part 1: theory and DGP

i. Solving for the equilibria

In scenario 1, the firm solves

$$\max_{L_t} p_t e^{\nu_t} L_t^\beta - w_t L_t,$$

with FOC

$$w_t = \beta p_t e^{\nu_t} L_t^{\beta-1},$$

which is the labor demand equation in this case. In scenario 2, where the firm knows it will affect wages, we also have to incorporate the $\partial w_t / \partial L_t$ term from the labor supply. The FOC in scenario 2 is

$$\beta p_t e^{\nu_t} L_t^{\beta-1} = w_t + L_t \frac{\partial w_t}{\partial L_t}.$$

We can already see that wages will be lower in this case. Because labor supply is upward-sloping in wages, the equation above will yield the wages of scenario 1 minus a positive term. Taking logs in the labor supply equation, we get

$$\log(L_t) = \alpha + \gamma \log(w_t) + \mu \log(u_t) + \rho \log(u_t) \log(w_t) + \eta_t.$$

This makes it easy to calculate the labor supply elasticity

$$\frac{\partial \log(L_t)}{\partial \log(w_t)} = \frac{w_t}{L_t} \frac{\partial L_t}{\partial w_t} = \gamma + \rho \log(u_t).$$

By the inverse function theorem,

$$\frac{\partial \log(w_t)}{\partial \log(L_t)} = \frac{L_t}{w_t} \frac{\partial w_t}{\partial L_t} = \frac{1}{\gamma + \rho \log(u_t)}.$$

Dividing both sides of the scenario 2 labor demand equation by w_t ,

$$\frac{\beta p_t e^{\nu_t} L_t^{\beta-1}}{w_t} = 1 + \frac{L_t}{w_t} \frac{\partial w_t}{\partial L_t} = 1 + \frac{1}{\gamma + \rho \log(u_t)}.$$

Solving for w_t ,

$$w_t = \beta p_t e^{\nu_t} L_t^{\beta-1} \quad (\text{Scenario 1 Labor Demand})$$

$$w_t = \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1} \beta p_t e^{\nu_t} L_t^{\beta-1}. \quad (\text{Scenario 2 Labor Demand})$$

We can combine both by defining the markdown

$$M = \begin{cases} 1, & \text{in Scenario 1} \\ \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1}, & \text{in Scenario 2} \end{cases}.$$

Then, in both cases, labor market equilibrium (L_t, w_t) is determined by both labor supply and demand, so by the system

$$\begin{cases} w_t = \beta M p_t e^{\nu_t} L_t^{\beta-1} \\ L_t = \exp\left(\alpha + \gamma \log(w_t) + \mu \log(u_t) + \rho \log(u_t) \log(w_t) + \eta_t\right) \end{cases}.$$

If we knew all the parameters $(\beta, \alpha, \gamma, \mu, \rho)$, and we observed $(p_t, u_t, \nu_t, \eta_t)$, we would also know M . Hence, the only unknowns in the two equations would be (L_t, w_t) . We could solve for them by hand by substituting w_t from the labor demand into the labor supply equation, from which we could find a root L_t^* . Then, we just plug it into either equation to compute w_t^* .

1. Wages in Scenario 1 vs. in Scenario 2

Because the markdown term M lies in $(0, 1)$, the labor demand curve is shifted down in Scenario 2. Therefore, equilibrium wages will be lower in Scenario 2.

2. Markdown

By the markdown formula

$$M = \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1},$$

M is increasing in γ , in ρ , and does not depend on μ . It is also increasing in t . Because M is a multiplier, M becoming larger means that wages are being marked down less, hence higher. Here, we rely on $u_t > 1$ to avoid the logs being negative, so unemployment u_t would be defined in absolute terms and not as a fraction.

II. Demand systems for bikes?

Part 1: theory and DGP

i. Derive expression for share of consumers

Each consumer i , when consuming bike j , obtains utility

$$U_{ij} = \exp(\beta_w w_j + \beta_g g_j) - \alpha_i p_j,$$

where $\alpha_i \stackrel{i.i.d.}{\sim} \text{LogNormal}(\mu, 1)$

The probability that consumer i chooses bike j is typically

$$P_{ij} = P\{U_{ij} \geq U_{ik} \quad \forall k\}.$$

But, because there is no consumer-good idiosyncratic term, there can be ties with non-zero probability. This is very similar to the Bresnahan vertical differentiation model, but with heterogeneity in α_i that multiplies prices, instead of the v_i that multiplies qualities. To deal with this, we will reduce the good characteristics into a single measure of quality, which is all that matters for decision-making. Define the quality of good j as

$$q_j = \exp(\beta_w w_j + \beta_g g_j).$$

Then the utility obtained when consuming good j is $U_{ij} = q_j - \alpha_i p_j$. The consumer chooses from a menu of (q_j, p_j) to maximize utility, depending on α_i . Without loss of generality, we can reorder goods using $q_1 \leq q_2 \leq \dots \leq q_J$, with the possibility of ties.

If there is a tie between goods j and $j+1$, where $q_j = q_{j+1}$, one of two things will occur. The first case is $p_j = p_{j+1}$, and both are equally good choices for the consumer. In this case, we assume the consumer will flip a coin to pick each with 50% probability. If there is a perfect tie between k goods, the consumer may pick each with probability $1/k$. The second case is that $p_j \neq p_{j+1}$, so we can assume $p_j < p_{j+1}$ WLOG. But then, $U_{i,j} > U_{i,j+1}$ for all consumers i , meaning that good $j+1$ is strictly dominated. As no one will choose it, it will have zero market share. Our approach is that, in case of a perfect tie, we remove all but one of the options from the choice set. The remaining good is representative of all duplicates combined.

In general, any good k is strictly dominated by j if $(q_k, -p_k) \leq (q_j, p_j)$, element-by-element, and they are not exactly equal. We can safely remove strictly dominated goods from our menu. Since we are removing ties, this already implies that both qualities and prices much go in the same direction. We can arrange the remaining goods such that $q_1 < q_2 < \dots$ and $p_1 < p_2 < \dots$. Even more strongly, assume three consecutive goods j , $j+1$, and $j+2$, have

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

We will show that almost no consumer will pick good $j+1$. Assume consumer i chooses $j+1$. Then, he prefers it to both j and $j+2$, meaning $q_{j+1} - \alpha_i p_{j+1} \geq q_j - \alpha_i p_j$. Rearranging,

$$\alpha_i \leq \frac{q_{j+1} - q_j}{p_{j+1} - p_j}.$$

Doing the same for $j + 1$ and $j + 2$, as $q_{j+1} - \alpha_i p_{j+1} \geq q_{j+2} - \alpha_i p_{j+2}$, then

$$\alpha_i \geq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

Combining the two, this implies

$$\frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}} \leq \alpha_i \leq \frac{q_{j+1} - q_j}{p_{j+1} - p_j}.$$

Along with our initial assumption that the ratios are actually increasing, the only way both can be true is if α is exactly equal to both ratios. Because this happens with zero probability, removing goods like $j + 1$ also will not affect choice probabilities (almost everywhere), and will therefore not change market shares. This means that there is no loss to imposing the assumption

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} > \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

Therefore, we construct a reduced menu $\widetilde{C}$ of available pairs $(q_j, -p_j)$. Because we are only looking at a set of $(q, -p)$ choices and not individual goods, there are no perfect ties by construction. We also rule out strict dominance by removing goods j such that $(q_j, -p_j) \leq (q_k, -p_k)$ for some $k \neq j$. Goods are then placed in a strictly increasing order of both quality and prices. Finally, we remove the possibility of increasing slopes by, given this ordering, removing any good $j + 1$ such that

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

This is done iteratively: we order the goods and compute the slopes. Then, we remove any non-increasing slopes, reassign the order and repeat. When choosing from this menu, consumers will have a unique best choice with probability 1. To see why, assume a consumer is indifferent between $(q, -p)$ and $(q', -p')$. Then, $q - \alpha_i p = q' - \alpha_i p'$, which pins down a particular value for $\alpha_i = (q' - q)/(p' - p)$, which happens with zero probability, by the distribution of α_i . Also, the choice will never be empty: as they choose from a finite list of real numbers, one of them must be the highest. This means that, when integrating choice probabilities into market shares, we do not need to worry about multiplicity of choices.

Order the choices in $\widetilde{C}$ by quality, with $q_1 < q_2 < \dots < q_{\widetilde{J}}$. By construction, the same order applies to prices, $p_1 < p_2 < \dots < p_{\widetilde{J}}$, and to slopes. Finally, we can compute choice probabilities. Individual i chooses j if, and only if, $q_j - \alpha_i p_j \geq q_k - \alpha_i p_k$ for all pairs $(q_k, -p_k) \in \widetilde{C}$. But we can simplify the expression by only looking at neighbors, as in the Bresnahan vertical differentiation model, thanks to the decreasing slopes restriction.

Individual i prefers good j over $j + 1$ when $q_j - \alpha_i p_j \geq q_{j+1} - \alpha_i p_{j+1}$. Assume, by contradiction, that he prefers good $j + 2$ over j , so $q_j - \alpha_i p_j \leq q_{j+2} - \alpha_i p_{j+2}$. These two imply

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \alpha_i \leq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}},$$

Which is impossible in our menu $\widetilde{C}$ by construction. As a result, if a consumer prefers j over $j + 1$, they will also prefer j over $j + 2$, and so on. Symmetrically, if he prefers j over $j - 1$, he will also

prefer j over all lower goods. Because of this, preferring j to both neighbors is equivalent to j being the global optimum. The probability that individual i chooses j is

$$\begin{aligned} P_{ij} &= P\left\{i \text{ prefers } j \text{ to } j+1 \text{ and to } j-1\right\} \\ &= P\left\{q_j - \alpha_i p_j \geq q_{j+1} - \alpha_i p_{j+1}, \quad q_j - \alpha_i p_j \geq q_{j-1} - \alpha_i p_{j-1}\right\} \\ &= P\left\{\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \alpha_i \leq \frac{q_j - q_{j-1}}{p_j - p_{j-1}}\right\}. \end{aligned}$$

For a given consumer that observes his α_i , this expression is either 0 or 1. In knife-edge cases where a consumer falls exactly at the margin, this would predict that they could choose more than one good. But this occurs with zero probability, so when we aggregate, each pair $(q_j, -p_j)$ in $\tilde{C}$ will have market share

$$s(q_j, -p_j) = \int_0^\infty \mathbb{I}\left[\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \alpha \leq \frac{q_j - q_{j-1}}{p_j - p_{j-1}}\right] dF(\alpha) = F\left(\frac{q_j - q_{j-1}}{p_j - p_{j-1}}\right) - F\left(\frac{q_{j+1} - q_j}{p_{j+1} - p_j}\right),$$

where F is the cdf of α_i . Define ties_j equal to the number of goods exactly tied in both quality and price with good j , including j . If j has no ties, $\text{ties}_j = 1$. To summarize, the market share of a good j is

$$s_j = \frac{s(q_j, -p_j)}{\text{ties}_j}, \text{ if } (q_j, -p_j) \in \tilde{C}, \quad \text{and 0 otherwise.}$$

ii. Bertrand-Nash price equilibrium

On the supply side, we assume a static multi product Bertrand-Nash game. Alan and Bianchi choose prices simultaneously, each taking the rival firm's prices as given. Product characteristics and marginal costs are taken as given, and each firm chooses the prices of all products it owns to maximize its total profit.

Alan owns $\mathcal{J}_A = \{a_1, a_2, a_3\}$ and Bianchi owns $\mathcal{J}_B = \{b_1, b_2\}$. Firm f 's profit in market t is

$$\pi_{ft} = \sum_{k \in \mathcal{J}_f} (p_{kt} - mc_{kt}) s_{kt}(\mathbf{p}_t).$$

Assuming an interior equilibrium and differentiable demand, the FOC for each $j \in \mathcal{J}_f$ is

$$s_{jt} + \sum_{k \in \mathcal{J}_f} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0.$$

To write these FOCs compactly, define the ownership matrix

$$\Omega_{kj} = \begin{cases} 1, & \text{if products } k \text{ and } j \text{ are owned by the same firm,} \\ 0, & \text{otherwise.} \end{cases}$$

Here,

$$\Omega = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

Thus Ω simply tells the model which cross price effects a firm internalizes. Alan internalizes the effects of changing one Alan price on all three Alan products, while Bianchi does the same for its two products. Neither firm internalizes the effect of its pricing decision on the rival firm's profit.

We define

$$D_t = \left[\frac{\partial s_{kt}}{\partial p_{jt}} \right]_{k,j}$$

as the demand Jacobian. Then the five Bertrand first order conditions can be written as

$$\mathbf{s}_t + (\Omega \circ D_t)'(\mathbf{p}_t - \mathbf{m}\mathbf{c}_t) = \mathbf{0},$$

where $\circ$ is the elementwise product. Defining

$$\Delta_t = -(\Omega \circ D_t)',$$

gives

$$\mathbf{s}_t - \Delta_t(\mathbf{p}_t - \mathbf{m}\mathbf{c}_t) = \mathbf{0}.$$

in terms of matrices notation.

If we assume Δ_t is invertible,

$$\mathbf{p}_t - \mathbf{m}\mathbf{c}_t = \Delta_t^{-1}\mathbf{s}_t.$$

iii. Product positioning and marginal costs

We interpret w_t, g_t in the cost expression as the product characteristics w_j, g_j from the table (or we simply assume $w_{tj} = w_j, g_{tj} = g_j$, so that bike characteristics are fixed over time).

From above, we measure product quality by the deterministic utility component

$$\delta_j = \exp(\beta_w w_j + \beta_g g_j),$$

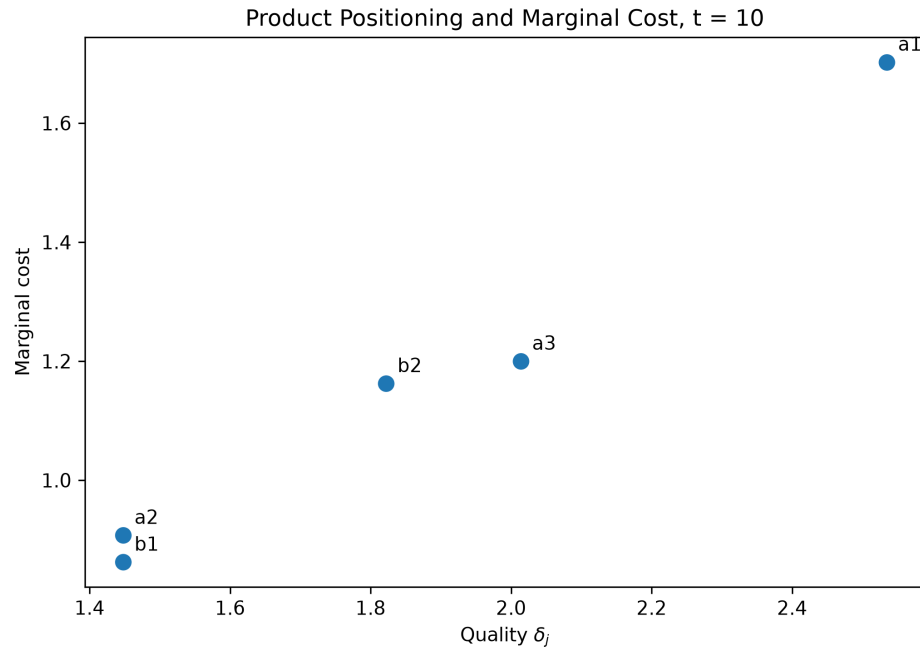
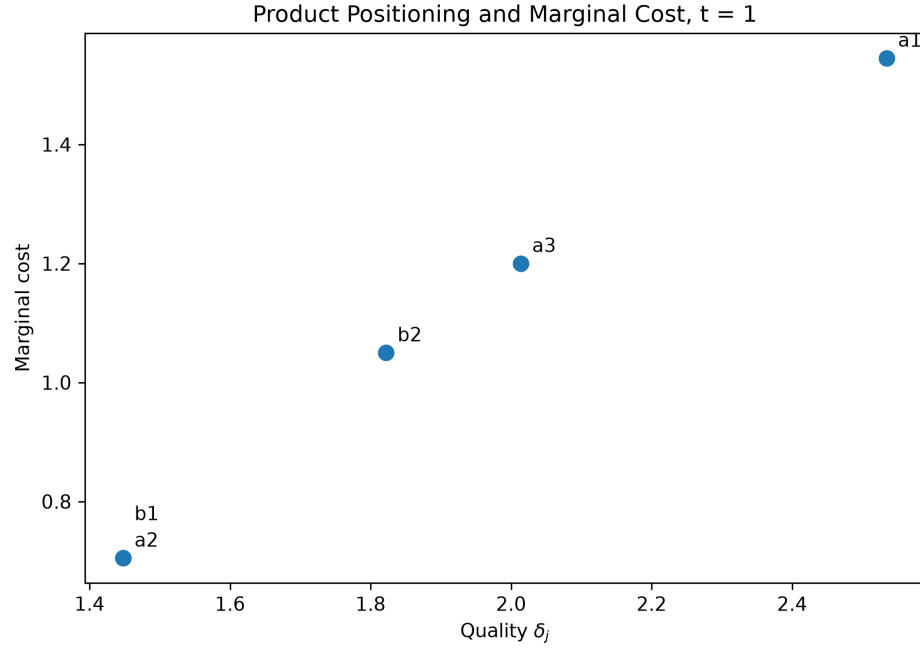
and marginal cost is, as defined by the question,

$$MC_{jt} = \gamma_w(1 - w_j)c_t + \gamma_g g_j, \quad c_t = 0.5 + \frac{t}{10}.$$

Thus

$$c_1 = 0.6, \quad c_{10} = 1.5.$$

The plots for $t = 1$ and $t = 10$ follows:



Note that for $t = 1$, a_2 and b_1 coincide as they have the same marginal cost. Also, we see that marginal cost rises over time for every product except a_3 , because $w_{a_3} = 1$ makes its carbon cost component equal to zero.

iv. Brief comment on market shares and competition

Product characteristics, and therefore δ_j , are fixed over time. The change in relative positioning therefore comes from marginal costs:

$$\Delta MC_j = \gamma_w(1 - w_j)(c_{10} - c_1) = 0.225(1 - w_j).$$

Hence,

$$(\Delta MC_{a_1}, \Delta MC_{a_2}, \Delta MC_{a_3}, \Delta MC_{b_1}, \Delta MC_{b_2}) = (0.1575, 0.2025, 0, 0.1575, 0.1125).$$

We therefore expect a_3 to gain market share because its marginal cost does not increase. b_2 should also gain relative to products whose costs rise more, while a_2 , which has the largest cost increase, should lose.

The comparison between a_2 and b_1 is clear as they have identical quality (same value on x-axis), but by $t = 10$,

$$MC_{a_2,10} = 0.9075 > 0.8625 = MC_{b_1,10},$$

so b_1 gains a cost advantage over a_2 .

The primitive demand side substitutability does not change because product characteristics are fixed. What changes is the firms' cost positioning. Competition between a_2 and b_1 becomes less symmetric as their costs separate, while on the other hand, the cost gap between a_3 and b_2 falls from 0.15 to 0.0375, making their pricing positions more similar and competitive pressure between them more likely to increase.

v.BNE for different μ

Please check the saved files.

vi. Coordination in a_2 and b_1 prices

Here Alan choose p_{a_1} and p_{a_3} to maximize π_A , Bianchi choose p_{b_2} to maximize π_B , and the pair (p_{a_2}, p_{b_1}) is chosen jointly to maximize total industry profit $\pi_A + \pi_B$ holding the other prices fixed. Thus the game has three price setting decision makers: Alan's noncoordinated products, Bianchi's noncoordinated product, and the coordinated (a_2, b_1) pair. We iterate these three best responses and verify that no decision maker has a profitable deviation. The agreement is valuable because $\delta_{a_2} = \delta_{b_1}$.

The two products have the same deterministic quality and, at $t = 1$, the same marginal cost, so they are very close substitutes. Under competition, cutting the price of one steals demand from the other. Coordinating their prices lets the firms take this effect into considerations and reduces the incentive to undercut each other.

Part 2: Estimation

i. Estimator

We define

$$\theta_d = (\mu, \beta_w, \beta_g).$$

The variance of $\log \alpha_i$ remains fixed at one, as specified in the DGP, so only μ is estimated from the distribution of price sensitivity.

For a given θ_d , predicted product shares are

$$s_{jt}(\theta_d) = \int \frac{\exp(\delta_j - \alpha p_{jt})}{\sum_k \exp(\delta_k - \alpha p_{kt})} f_\alpha(\alpha; \mu) d\alpha,$$

with same as above

$$\delta_j = \exp(\beta_w w_j + \beta_g g_j).$$

Because market size is normalized to one, the observed q_{jt} are market shares and satisfy $\sum_j q_{jt} = 1$. We therefore use them as choice frequencies. The multinomial log likelihood per consumer in market t is

$$\ell_t(\theta_d) = \sum_j q_{jt} \log s_{jt}(\theta_d).$$

The demand estimator is

$$\hat{\theta}_d = \arg \max_{\theta_d} \sum_{t=1}^{10} \ell_t(\theta_d).$$

Following the provided JAX structure, we write the likelihood for one market and use `vmap` to apply it to all 10 markets. For the supply side, we impose the competitive Bertrand ownership structure from Part 1(ii). Let

$$D_t = \left[\frac{\partial s_{kt}}{\partial p_{jt}} \right]_{k,j}$$

evaluated at $\hat{\theta}_d$ and the observed prices, and define

$$\hat{\Delta}_t = -(\Omega \circ D_t)'.$$

The Bertrand FOCs imply

$$\mathbf{q}_t - \hat{\Delta}_t(\mathbf{p}_t - \mathbf{mc}_t) = \mathbf{0},$$

so implied marginal costs are

$$\widehat{\mathbf{mc}}_t = \mathbf{p}_t - \hat{\Delta}_t^{-1} \mathbf{q}_t.$$

Finally, the structural cost equation is

$$MC_{jt} = \gamma_w(1 - w_j)c_t + \gamma_g g_j.$$

It contains no intercept, so we estimate (γ_w, γ_g) by OLS without a constant:

$$(\hat{\gamma}_w, \hat{\gamma}_g) = \arg \min_{\gamma_w, \gamma_g} \sum_{t,j} \left[\widehat{MC}_{jt} - \gamma_w(1 - w_j)c_t - \gamma_g g_j \right]^2.$$

ii. Estimation results

The estimates are

Dataset	$\hat{\mu}$	$\hat{\beta}_w$	$\hat{\beta}_g$	$\hat{\gamma}_w$	$\hat{\gamma}_g$
Competition 1	-0.5000	-0.1000	0.2000	0.2500	0.3000
Competition 2	0.0000	-0.1000	0.2000	0.2500	0.3000
Competition 3	0.5000	-0.1000	0.2000	0.2500	0.3000
Collusion 1	-0.5000	-0.1000	0.2000	1.6603	0.2106
Collusion 2	0.0000	-0.1000	0.2000	1.1025	0.2480
Collusion 3	0.5000	-0.1000	0.2000	0.7660	0.2724

For the competitive datasets, the estimator gets all five parameters almost exactly right, which is expected since the estimation model matches how these datasets were generated. For the collusive datasets, the demand estimates $\hat{\mu}$, $\hat{\beta}_w$, and $\hat{\beta}_g$ barely change. The reason is that collusion changes how firms set prices, but not how consumers choose among products. The cost estimates are different. Here the estimator still assumes Bertrand competition even though a_2 and b_1 were priced jointly, so part of the collusive markup is incorrectly absorbed into marginal cost. This is why $\hat{\gamma}_w$ is much larger than the true value 0.25. The distortion becomes smaller as μ increases: $\hat{\gamma}_w$ falls from about 1.66 to 0.77, while $\hat{\gamma}_g$ moves closer to 0.30. This pattern comes from these simulated datasets and is not a general feature of collusion.

Part 3: Extra and optional

i. Households choosing multiple bikes

Let household h have $n_h \in \{1, \dots, 5\}$ members. We assume that each member can buy at most one bike, so the household chooses a bundle

$$\mathbf{q}_h = (q_{ha_1}, \dots, q_{hb_2}), \quad q_{hj} \in \mathbb{Z}_+, \quad \sum_j q_{hj} \leq n_h.$$

We model the household's utility as the sum of the utilities from all bikes it buys:

$$U_h(\mathbf{q}) = \sum_j q_j \delta_j - \alpha_h \sum_j p_j q_j + \epsilon_{h\mathbf{q}},$$

where $\alpha_h \sim \text{LogNormal}(\mu, 1)$ and $\epsilon_{h\mathbf{q}}$ is i.i.d. Type-I extreme value across possible bundles. So, conditional on α_h , the household chooses among bundles using the same logit idea as before.

If λ_n is the fraction of households with n members, demand for product j is

$$Q_j(\mathbf{p}) = \sum_{n=1}^5 \lambda_n \sum_{\mathbf{q} \in \mathcal{B}_n} q_j P(\mathbf{q} \mid n).$$

The main difference from Part 1 is that Q_j is now the expected number of bikes purchased, rather than the fraction of consumers choosing product j , because one household can buy more than one bike.

On the supply side, we keep the same Bertrand pricing setup, but replace s_j and its price derivatives with Q_j and its derivatives.

For estimation, if we observe household size and the bundle each household chooses, we can estimate demand from those bundle choices, again integrating over α_h . We then recover marginal costs using the new quantity demand Jacobian and estimate (γ_w, γ_g) as in Part 2. If we only observe aggregate product quantities, we would instead match predicted and observed quantities using minimum distance or GMM.